STM32串口读取数据

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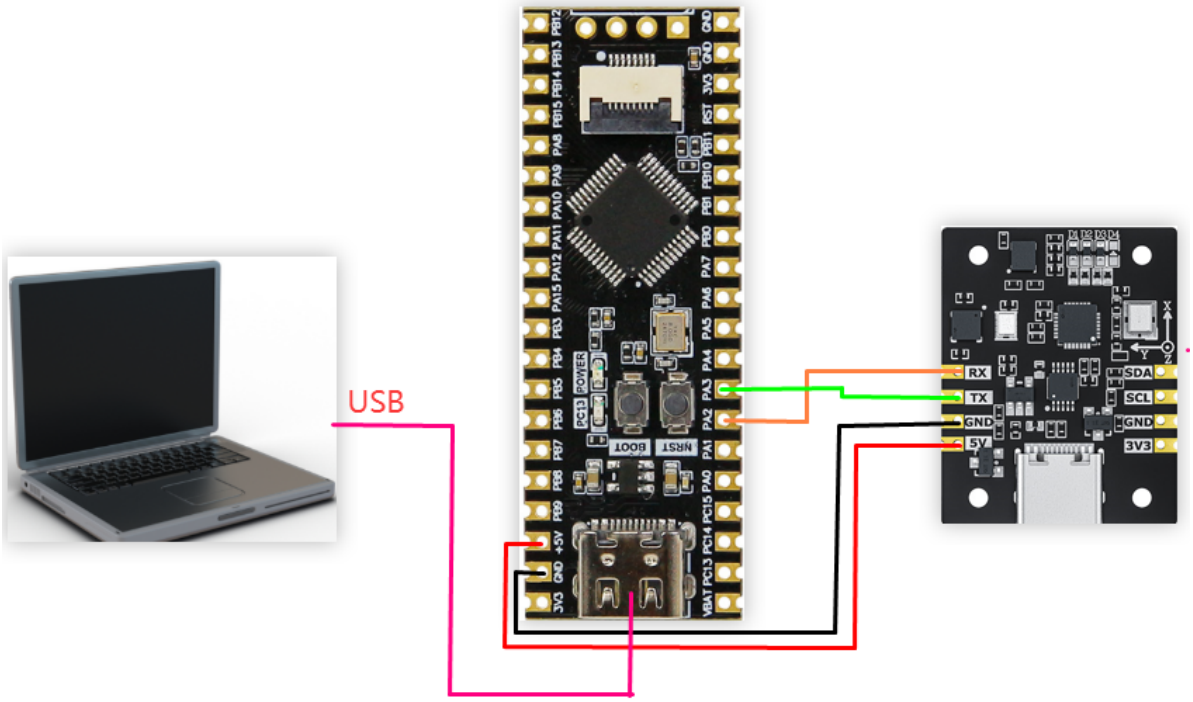
本次例程使用的是STM32F103C8T6核心板（亚博），一台windows电脑、杜邦线若干、IMU姿态传感器。

1.连接设备



IMU姿态传感器	STM32F103C8T6核心板（亚博）
RX	PA2
TX	PA3
GND	GND
5V	5V

2.关键代码解析

具体代码请看资料中的源码。

```
//解析环形缓冲中的数据，提取完整帧并更新缓存

//Process RX ring buffer, parse frames and update internal cache
```

```c
void IMU_UART_Process(void)
{
    enum {
        RX_STATE_EXPECT_HEAD1 = 0,
        RX_STATE_EXPECT_HEAD2,
        RX_STATE_EXPECT_LENGTH,
        RX_STATE_EXPECT_FUNCTION,
        RX_STATE_COLLECT_DATA
    };

    static uint8_t  rx_state = RX_STATE_EXPECT_HEAD1;
    static uint8_t  frame_length = 0;
    static uint8_t  frame_function = 0;
    static uint8_t  frame_buffer[64]; /* 数据区 + 校验 / data section + checksum
*/
    static uint16_t frame_index = 0;

    uint8_t current_byte = 0;

    while (_rxbuf_pop(&current_byte) == 0) {
        switch (rx_state) {
        case RX_STATE_EXPECT_HEAD1:
            rx_state = (current_byte == FRAME_HEAD1) ? RX_STATE_EXPECT_HEAD2 :
RX_STATE_EXPECT_HEAD1;
            break;

        case RX_STATE_EXPECT_HEAD2:
            rx_state = (current_byte == FRAME_HEAD2) ? RX_STATE_EXPECT_LENGTH :
RX_STATE_EXPECT_HEAD1;
            break;

        case RX_STATE_EXPECT_LENGTH:
            frame_length = current_byte;
            rx_state = RX_STATE_EXPECT_FUNCTION;
            break;

        case RX_STATE_EXPECT_FUNCTION:
            frame_function = current_byte;
            frame_index = 0;
            rx_state = RX_STATE_COLLECT_DATA;
            break;

        case RX_STATE_COLLECT_DATA: {
            uint16_t data_length = (frame_length >= 4) ? (uint16_t)(frame_length
- 4) : 0;
            if (data_length == 0 || data_length > sizeof(frame_buffer)) {
                rx_state = RX_STATE_EXPECT_HEAD1;
                break;
            }

            frame_buffer[frame_index++] = current_byte;
            if (frame_index >= data_length) {
                uint8_t calculated_checksum = (uint8_t)(FRAME_HEAD1 +
FRAME_HEAD2 + frame_length + frame_function);
                for (uint16_t i = 0; i < data_length - 1; ++i) {
                    calculated_checksum = (uint8_t)(calculated_checksum +
frame_buffer[i]);
                }
```

```
                uint8_t received_checksum = frame_buffer[data_length - 1];
                if (calculated_checksum == received_checksum) {
                    _parse_frame_data(frame_function, frame_buffer);
                }
                rx_state = RX_STATE_EXPECT_HEAD1;
            }
        } break;

        default:
            rx_state = RX_STATE_EXPECT_HEAD1;
            break;
        }
    }
}


/* ---------- 解析数据帧 / Parse one complete frame ---------- */
static void _parse_frame_data(uint8_t frame_function, const uint8_t *frame_data)
{
    if (frame_function == IMU_FUNC_RAW_ACCEL) {
        float accel_ratio = 16.0f / 32767.0f;
        s_ax = to_int16(&frame_data[0])  * accel_ratio;
        s_ay = to_int16(&frame_data[2])  * accel_ratio;
        s_az = to_int16(&frame_data[4])  * accel_ratio;

        float deg_to_rad = 3.14159265358979323846f / 180.0f;
        float gyro_ratio  = (2000.0f / 32767.0f) * deg_to_rad;
        s_gx = to_int16(&frame_data[6])  * gyro_ratio;
        s_gy = to_int16(&frame_data[8])  * gyro_ratio;
        s_gz = to_int16(&frame_data[10]) * gyro_ratio;

        float mag_ratio = 800.0f / 32767.0f;
        s_mx = to_int16(&frame_data[12]) * mag_ratio;
        s_my = to_int16(&frame_data[14]) * mag_ratio;
        s_mz = to_int16(&frame_data[16]) * mag_ratio;
    } else if (frame_function == IMU_FUNC_EULER) {
        s_roll  = to_float(&frame_data[0]);
        s_pitch = to_float(&frame_data[4]);
        s_yaw   = to_float(&frame_data[8]);
    } else if (frame_function == IMU_FUNC_QUAT) {
        s_q0 = to_float(&frame_data[0]);
        s_q1 = to_float(&frame_data[4]);
        s_q2 = to_float(&frame_data[8]);
        s_q3 = to_float(&frame_data[12]);
    } else if (frame_function == IMU_FUNC_BARO) {
        s_height            = to_float(&frame_data[0]);
        s_temperature       = to_float(&frame_data[4]);
        s_pressure          = to_float(&frame_data[8]);
        s_pressure_contrast = to_float(&frame_data[12]);
    } else if (frame_function == IMU_FUNC_VERSION) {
        s_version_high = frame_data[0];
        s_version_mid  = frame_data[1];
        s_version_low  = frame_data[2];
    } else if (frame_function == IMU_FUNC_RETURN_STATE) {
        s_last_rx_function = frame_data[0];
        s_last_rx_state    = (int16_t)frame_data[1];
    }
```

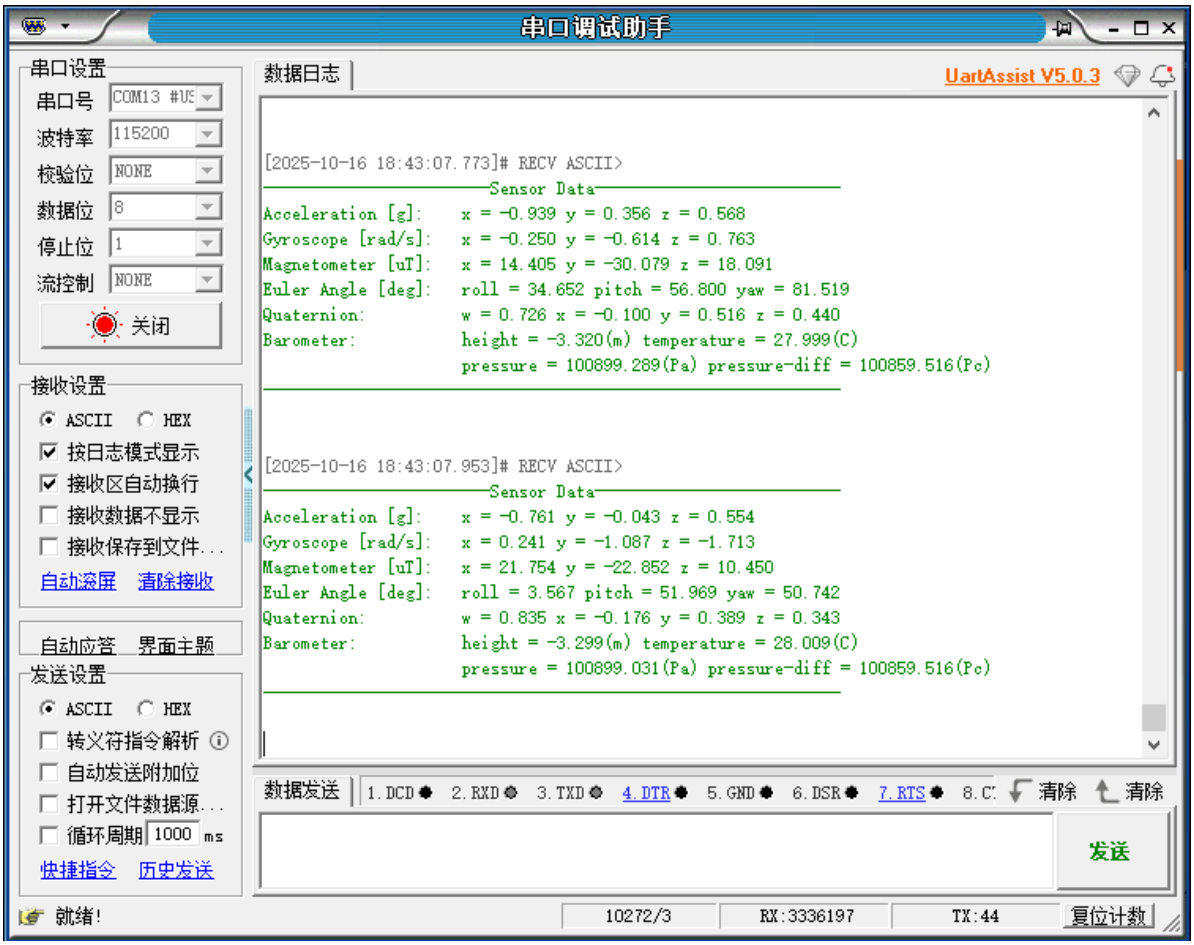
```
}
```

IMU_UART_Process(): 读取缓存的数据，并调用_parse_frame_data解析符合通信协议的数据。

_parse_frame_data()：解析数据。

3.读取imu数据

程序下载进入STM32后，打开串口助手（配置参数如下图所示），可以看到一直打印IMU模块的数据，当我们改变IMU模块的姿态，数据会发生变化。



注意：以上为10轴IMU的数据读取，6轴无磁力计（Magnetometer）与气压计（Barometer）数据，9轴无气压计（Barometer）数据。